

Homework — 2.1.3 Thinking Procedurally — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] — Two correct matches per mark:

Term	Match
Decomposition	A
Sub-procedure	D
Dependency	E
Independent steps	B

Distractors not used: C (abstraction) and F (concurrent/parallel processing).

2. [3] — Two correctly numbered steps per mark:

Step	Order
Lock door	1
Fill with water	2
Heat water	3
Wash	4
Drain	5
Spin	6

Each step needs the state produced by the one before it (water must be in the drum before heating, heated before washing, drained before spinning), so this is the only valid order.

3. [4] — 1 mark per row; for a false row the mark needs **False** plus a valid correction.

Statement	True / False	Correction
issueTicket could move before takePayment	False	issueTicket uses the output of takePayment (confirmation of successful payment), so it must stay after it — the order is fixed.
Decomposition lets team members build modules at the same time	True	—
The countdown-display update must always run after the wash step	False	The display update and the wash step do not use each other's outputs — there is no dependency, so their relative order is not fixed.
A sub-procedure means a fix is made in one place	True	—

4. [2] — Two correct gaps per mark: (a) **sub-problem**; (b) **output**; (c) **dependency**; (d) **fixed**.
(Distractors: input, abstraction, concurrent.)

5. [3] — Any three plausible sub-procedures (1 each): browse/select film and showing; check seat availability; allow the user to choose seats; take/validate payment; issue/email the ticket; update the seat-availability records; create the booking record. Must be plausible sub-tasks of the booking scenario — generic answers ("do the input", "do the processing") earn nothing.

Part B (6 marks)

6. [2] — 1 mark: a precondition is a condition that must be **true before** a routine runs for the routine to work correctly. 1 mark: for binary search, the list must already be **sorted** (accept: the list is not empty).

7. [2] — 1 mark each, in the correct order: **lexical analysis** (tokenising) then **syntax analysis** (parsing).
(Code generation and optimisation follow.)

8. [2] — 1 mark + 1 mark for one developed advantage: Agile is **iterative**, delivering working software in increments and accommodating changing requirements [1]; the customer sees progress early and changes are cheaper to make, whereas Waterfall fixes requirements up front so late changes are costly [1].

Total: 14 + 6 = 20 marks